Preferences and indifference curves

Lecture notes

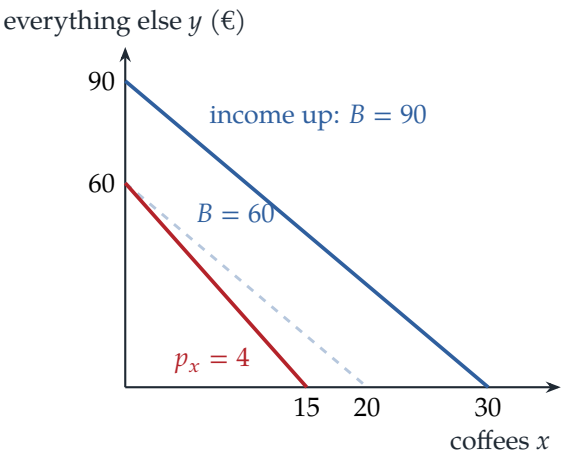
What this lecture is about. Last lecture described what a consumer *can* buy: the budget set. Today we describe what a consumer *wants*. We do this without any formula. We start from four simple rules about how people rank bundles, and from those rules we draw the indifference curves. They show how a consumer trades one good against another. Then we ask what we can learn about preferences just by watching choices on budget lines.

Contents. 1 The budget set, briefly · 2 Preferences and four rules · 3 Indifference curves · 4 Trade-offs: the marginal rate of substitution · 5 Perfect substitutes and perfect complements · 6 Corners, a first look · 7 What choices reveal · 8 Next lecture

1 The budget set, briefly

A simple example. Each month you spend a budget of $B = €60$ on coffees (x, at $p_x = €3$ each) and on everything else (y, the composite good, measured in euro, so $p_y = 1$). The budget line is

$$3x + y = 60 \qquad \Longleftrightarrow \qquad y = 60 - 3x.$$



- **The two ends.** All of the budget on coffees gives $B/p_x = 60/3 = 20$ coffees. All of it on everything else gives $B/p_y = €60$.
- **Income rises** to $B = €90$: both ends move out (30 coffees, €90), the slope stays -3. The line *shifts* out, parallel to itself.
- **The coffee price rises** to $p_x = €4$: the x-end moves in to $60/4 = 15$, the y-end stays at €60. The line *rotates* around the y-end and becomes steeper.
- **The composite good.** Because y is money for everything else, $p_y = 1$. The y-end is then the budget itself, and the slope is simply $-p_x$: each coffee costs €3 of everything else.

What is the slope of the budget line called?

The size of the slope, p_x/p_y, is the **relative price** of x: how many units of y the *market* makes you give up for one more unit of x. It is the opportunity cost of x in units of y. Some books call it the *market rate of exchange*.

It is **not** the same as how much you would be *willing* to give up; that is a different idea (lecture 8, on trade-offs).

2 Preferences and four rules

2.1 Bundles and how we compare them

A **bundle** $A = (x_A, y_A)$ lists an amount of each good, for example 8 coffees and €36 of everything else. A consumer's **preferences** are her ranking of all bundles. We only ask which of two bundles she likes more; we never ask *how much* more.

Notation

$A \succsim B$	A is **at least as good as** B	the basic relation ("weakly preferred")
$A \succ B$	A is **strictly preferred** to B	$A \succsim B$, but not $B \succsim A$
$A \sim B$	the consumer is **indifferent**	$A \succsim B$ and $B \succsim A$

Everything starts from the single symbol $\succsim$. Strict preference and indifference are both defined from it.

Economists assume that preferences follow four rules. The first two are about consistency. The last two are about the goods themselves.

2.2 Rule 1: completeness

Completeness

For any two bundles A and B, at least one of $A \succsim B$ or $B \succsim A$ holds. The consumer can always compare.

"I can't decide" is not allowed. Note the difference with indifference: an indifferent consumer *has* compared the two bundles and found them equally good. Completeness rules out the consumer who refuses to compare at all.

2.3 Rule 2: transitivity

Transitivity

If $A \succsim B$ and $B \succsim C$, then $A \succsim C$.

Transitivity says the ranking hangs together: it has no cycles. It looks obvious, but plenty of everyday relations are not transitive. Team 1 can beat team 2 and team 2 can beat team 3, while team 3 still beats team 1. Your friend's friend need not be your friend.

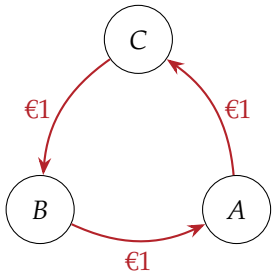
Example: the money pump

Suppose you rank three bundles A, B and C in a cycle:

$$A \succ B, \qquad B \succ C, \qquad C \succ A.$$

You hold bundle C. A seller offers to switch you to B for a fee of €1. You prefer B to C, so you pay. Then the seller offers A for another €1: you prefer A to B, so you pay again. Then the seller offers C for €1: you prefer C to A, so you pay a third time.

After three swaps you hold exactly the bundle you started with, and you are €3 poorer. The seller can repeat this forever. A person with cyclic preferences can be "pumped" until the money is gone. This is the classic argument for transitivity: people with intransitive preferences would lose money to anyone who noticed.



each arrow: a swap you are happy to pay for

Insight: where intransitivity comes from in real life

Real people do sometimes choose in cycles. A common reason:

- **Many considerations, counted like votes.** If you judge options on price, quality and speed, and pick the option that wins on two of the three, you can end up in a cycle even though each single consideration ranks the options consistently. The same happens to groups: three voters, each with a perfectly transitive ranking, can produce a cycle under pairwise majority voting.

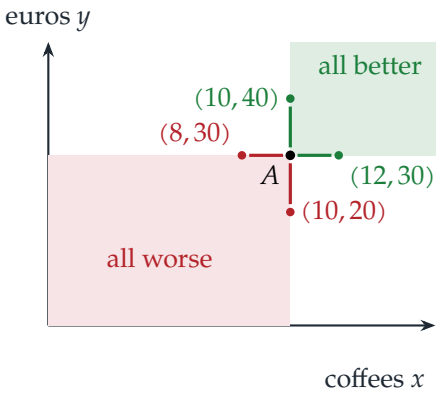
The theory keeps transitivity as an assumption. Section 7 shows how we can *test* it with data.

2.4 Rule 3: more is better

More is better (monotonicity)

If bundle A has at least as much of each good as B, and more of at least one, then $A \succ B$.

Start at $A = (10, 30)$: 10 coffees and €30. Two more coffees and the same money, $(12, 30)$, is better. The same coffees and €10 more, $(10, 40)$, is better too. So is everything up and to the right of A: more of one good and no less of the other. In the same way $(8, 30)$ and $(10, 20)$ are worse, and so is everything down and to the left.



This rule is about *goods*: things you want. It fails for *bads*, such as noise, pollution or spam messages, where more is worse. With a bad, the rule must be turned around (less of the bad is better). It can also fail when you are full (satiated): ten slices of pizza may be worse than nine. In this course both goods are goods unless we say otherwise.

2.5 Rule 4: convexity

Convexity

Take two equally good bundles, $A \sim B$. Then any mix of them is at least as good as either one:

$$tA + (1 - t)B \succsim A \qquad \text{for every } t \text{ between 0 and 1.}$$

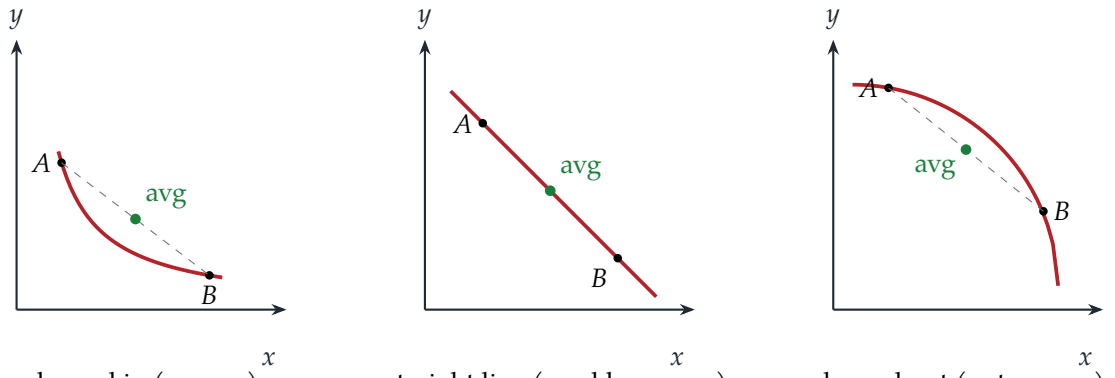
A **mix** takes a share t of bundle A and a share $1 - t$ of bundle B. The half-half mix ($t = \frac{1}{2}$) is simply the **average** of the two bundles. If the mix is always *strictly* better (for every t strictly between 0 and 1, with $A \neq B$), preferences are **strictly convex**.

Averages beat extremes. Suppose you are indifferent between a week of eating only pizza and a week of eating only salad. Convexity says that a week with half pizza and half salad is at least as good as either extreme. Most people like variety: the first salad after six pizza days is worth a lot.

The same idea in numbers. Take two bundles you like equally: $A = (2, 50)$, 2 coffees and €50 of everything else, and $B = (14, 14)$. Their average is

$$\tfrac{1}{2}A + \tfrac{1}{2}B = \left(\tfrac{2+14}{2}, \tfrac{50+14}{2}\right) = (8, 32).$$

Convexity says that $(8, 32)$ is at least as good as A and B. In a picture, the average lies at the midpoint of the straight segment between A and B.



bowed in (convex): the average is *better*

straight line (weakly convex): the average is *equally good*

bowed out (not convex): the average is *worse*

Insight: does the curve have to be bowed in? No

Convexity is an *assumption*, not a law of nature. Three things are worth knowing.

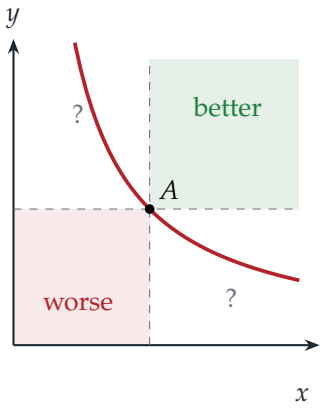
1. **A straight line is allowed.** The definition says the mix is *at least as good*. On a straight curve of equally good bundles the mix lies on the same curve, so it is exactly as good. That satisfies (weak) convexity. Only *strict* convexity fails. Perfect substitutes (lecture 8) have straight-line curves and are weakly convex.
2. **What convexity really rules out** is a curve bowed *away* from the origin. There the extremes beat the average: you would always want to spend everything on one good. That does happen: you may be indifferent between painting your room all red and all blue, but a half-half mix gives a purple you dislike: worse than either.
3. **Why we like the assumption.** Convexity is what makes the analysis simple. Section 6, on corners, shows why, once the budget line is back next to these curves.

3 Indifference curves

3.1 Building the curve from the rules

Pick any bundle A. The rules already tell us a lot about which bundles are better or worse than A.

- Every bundle up and to the right of A has more of at least one good and no less of the other: by *more is better*, it is **better** than A.
- Every bundle down and to the left of A is **worse** than A, for the same reason.
- So the bundles that are exactly as good as A can only lie in the other two areas: up and to the left (more y, less x), or down and to the right (more x, less y).



Join all bundles that are exactly as good as A. The result is the **indifference curve** through A, and it must slope downward.

Indifference curve

An **indifference curve** is the set of all bundles that the consumer likes exactly as much as a given bundle. A family of such curves, one through every bundle, is the **indifference map**. Curves further from the origin are better.

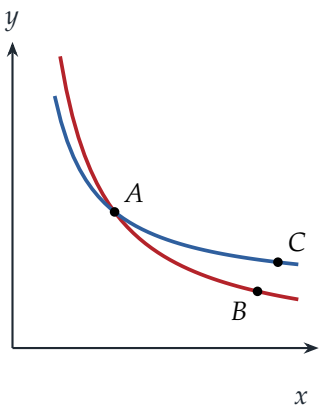
3.2 Five properties, and which rule gives each

property of indifference curves	comes from
1. there is a curve through every bundle	completeness
2. curves slope downward	more is better
3. curves further out are better	more is better
4. curves never cross	transitivity *and* more is better
5. curves are bowed toward the origin	convexity

An upward-sloping curve? If an indifference curve sloped upward, a bundle further up the curve would have more of *both* goods and yet be only equally good. That breaks *more is better*, provided both are goods. (If y were a bad, such as minutes of commuting, an upward slope makes sense: you need more coffees to put up with a longer commute.)

3.3 Why indifference curves cannot cross

Suppose two curves cross at bundle A, as in the figure. Bundle B is on the first curve, bundle C on the second, and C lies up and to the right of B.

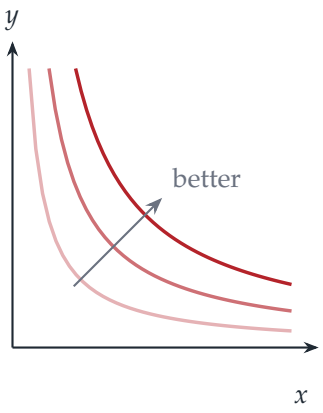


- $A \sim B$ (same curve), and $A \sim C$ (same curve).
- By *transitivity*, $B \sim C$.
- But C has more of both goods than B (it lies up and to the right), so by *more is better*, $C \succ B$.

Both cannot be true. So two indifference curves can never cross. Notice that we needed *two* rules for this argument.

3.4 The indifference map

Every bundle has its own curve, so the plane is filled with curves, like the contour lines of a hiking map. Picture utility as a mountain over the floor of bundles: a flat plane that cuts the mountain at one height traces one indifference curve, and seen from above the cuts at several heights are the contour lines $u_1 < u_2 < u_3$. Each curve is a "height" of satisfaction. Moving to a curve further out means moving to a better bundle. The map is a complete picture of the consumer's preferences. Two people with different tastes have different maps: someone who loves coffee has steep curves (she gives up a lot of other spending for one more coffee), someone who cares little about them has flat curves. In numbers: on the coffee lover's curve, going from 9 coffees and €13 to 10 coffees and €10 means giving up €3 for the 10th coffee. On the other person's curve, going from 9 coffees and €10.50 to 10 coffees and €10, the 10th coffee costs only €0.50. Both end at the same bundle $(10, €10)$, yet the 10th coffee is worth €3 to one and €0.50 to the other: that difference is the steepness.



4 Next lecture

Lecture 8 continues from here: the marginal rate of substitution (how much of one good you would give up for one more unit of the other), perfect substitutes and perfect complements, the best feasible bundle, and utility functions.

Summary

- The budget line's slope, p_x/p_y, is the relative price: the market's rate of exchange.
- Preferences rank bundles: $\succsim$ (at least as good), $\succ$ (strictly better), $\sim$ (indifferent).
- Four rules: completeness, transitivity (no cycles; otherwise a money pump), more is better (for goods), convexity (averages at least as good as extremes).
- An indifference curve joins equally good bundles. Curves slope down and further out is better (more is better), never cross (transitivity and more is better), and bow in (convexity).

Further reading. Frank, *Microeconomics and Behavior*, chapter 3, pp. 63–68 (week 4 page, additional readings; the password is given in class).